A Grounded Typology of Word Classes

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Abstract

We propose a grounded approach to meaning in language typology. We treat data from perceptual modalities, such as images, as a language-agnostic representation of meaning. Hence, we can quantify the function–form relationship between images and captions across languages. Inspired by information theory, we define "groundedness", an empirical measure of contextual semantic contentfulness (formulated as a difference in surprisal) which can be computed with multilingual multimodal language models. As a proof of concept, we apply this measure to the typology of word classes. Our measure captures the contentfulness asymmetry between functional (grammatical) and lexical (content) classes across languages, but contradicts the view that functional classes do not convey content. Moreover, we find universal trends in the hierarchy of groundedness (e.g., nouns > adjectives > verbs), and show that our measure partly correlates with psycholinguistic concreteness norms in English. We release a dataset of groundedness scores for 30 languages. Our results suggest that the grounded typology approach can provide quantitative evidence about semantic function in language.

1 Introduction

Within linguistics, *typology* is the subfield focused on the study of patterns and variation across the world's languages (Croft, 2002, pp. 1–2). To identify such patterns, linguists must carefully identify phenomena of interest within languages, and then align them with one another. For example, vowels exist in a continuous acoustic and perceptual space, without clear boundaries between them. To define vowel categories and align systems across languages, linguists rely largely on acoustic properties of the speech signal—reducing the problem to a physically grounded, empirical one (Liljencrants et al., 1972; Cotterell and Eisner, 2017).

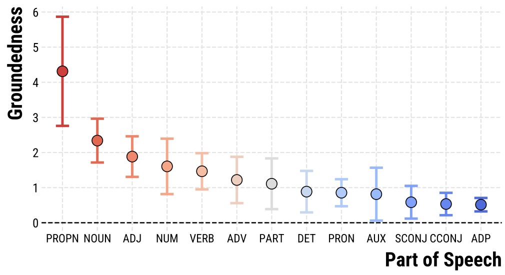

Figure 1: Mean and standard deviation of per-language mutual information estimates between word class and image. Across 30 languages, we see clear and consistent tendencies about which parts of speech are more "grounded", corresponding to a distinction between lexical and functional classes.

While empirically grounding language form (surface structure like vowels) is typically straightforward, language is not just a formal system, but also a functional one. Many questions within typology relate to the relationship between form and *meaning*, especially in domains like morphology and syntax. Typically, typologists manually identify semantic/functional roles such as "subject", and "causative" and study their expression across languages (Haspelmath, 2010; Greenberg, 1966). Unlike with many definitions based on form, definitions based on meaning are left up to subjective discretion, leading to debates which reduce to the definition of particular terms cross-linguistically (Haspelmath, 2007, 2012; Plank, 1994).

Instead, we propose a "grounded" approach to typology, which (under certain assumptions), allows the quantification and cross-linguistic comparison of language function and semantics across languages. By looking at sentences produced as captions of the same image across languages, we can use the image as an evidence-based, language-agnostic representation of the shared semantics underlying these utterances, similar to the evidence-based acoustic signal in the study of vowel spaces.

In this work, we specifically focus on semantic contentfulness—how semantically informative a given word token is. We introduce a way to empirically quantify contentfulness, *groundedness*, which relies on vision-and-language models. Groundedness measures how much less surprising a word is when we know the perceptual stimuli (i.e., the image) it describes. This *surprisal difference* between the surprisal of the word token in an image captioning model versus its surprisal in a language model is an estimate of the pointwise mutual information: the greater this difference (LM > captioning), the more *grounded* the word is in that context.

As a case study, we apply this measure to the study of the typology of word classes ("parts of speech"). Literature from cognitive, pyscho- and neurolinguistics all point to contentfulness being an organizing factor in word class processing and even formation and structure: low-content (functional) word classes have many different properties from high-content (lexical) classes (Dubé et al., 2014; Bird et al., 2003; Chiarello et al., 1999). Yet, there has been no cross-linguistic study of the relationship between contentfulness and word class.

Using our groundedness measure to quantify semantic contentfulness, we can estimate the mutual information of a word class with a caption's meaning (image). We find our measure largely rediscovers the distinction between lexical and functional word classes across 30 languages. Further, though it correlates only weakly with psycholinguistic norms for imageability and concreteness in English, it provides an intuitive ranking (noun > adjectives > verbs) across languages. On the other hand, it contradicts the view of adpositions as a "semi-lexical" class (Corver and Riemsdijk, 2001) and suggests grammatical word classes do carry some semantic content. These results thus partly validate and partly falsify received wisdom about word class contentfulness. They suggest the utility of this measure as a general tool for studying contentfulness in linguistics, and of taking a grounded approach to typological problems. We release the model used to estimate our measure and a dataset of groundedness values in 30 languages.[1]

2 Background

An excellent example of the relevance of the relationship between semantic function and linguistic form to typology is *word classes*. Within a particular language, there are typically groups of words unified by the (formal) contexts in which they can appear. Further, this distribution of words is not arbitrary, but unified by a particular semantic prototype. For example, in English, nouns are a class of words which prototypically denote physical objects or things and can follow words like "*the*", "*this*", and "*that*". However, not all languages have words like "*the*", and so an equivalent formal–structural criterion cannot be given (Haspelmath, 2012). On the other hand, semantic criteria are not sufficient to describe these classes: most languages can express prototypical verb or adjective meanings with the syntactic distribution of a noun.

The elusiveness of a cross-linguistic definition for word classes leads to many debates about particular languages "having" or "not having" a distinction between (e.g.) nouns and verbs on the basis of a mix of formal and semantic criteria (cf. Kaufman, 2009; Hsieh, 2019; Richards, 2009; Weber, 1983; Floyd, 2011). Here, we investigate word classes as operationalized in a framework where there is a fixed set of *universally applicable* word classes, as set out in the Universal Dependencies project (de Marneffe et al., 2021).While this is problematic in general, our aim is not to claim that the assignment of word classes is precisely correct, but rather to empirically and quantitatively investigate the functional/semantic dimension of this common operationalisation of word class. In future work, we aim to investigate the relationship between these measures and non-prototypical parts of speech.

2.1 Contentfulness and word class

In this work, we focus on the related distinction between lexical/contentful word classes (e.g. nouns, verbs, and adjectives) and functional/grammatical word classes. Functional word classes are typically closed-class, meaning they do not admit new members and typically do not exhibit rich productive morphology; they tend to express highly grammatical and abstract meanings. Lexical classes are typically open class, productively admitting new members, and their meanings tend to be more concrete and contentful (Corver and Riemsdijk, 2001).

Complications about these generalized categories and tendencies abound, however. For example, in some languages like Jaminjung, prototypically lexical categories like verbs are closed class (Schultze-Berndt, 2000; Pawley, 2006). Further, both the abstraction and semantic contentfulness of particular members of a given word class can be quite variable. For example, a noun like "*factor*"

[1] https://osf.io/bdhna/

has a highly abstract meaning, while the meaning of the preposition "*to*" is intuitively more abstract than the preposition "*above*", despite belonging to the same, "abstract" grammatical word class. Further, over time words can change in both their contentfulness and even word class through processes like grammaticalization (Bisang, 2017).

Nevertheless, the complex relationship between contentfulness and word class remains unexplored through a cross-linguistic empirical lens—perhaps due to the difficulties of measuring such properties.

2.2 Measuring contentfulness

The relationship between contentfulness and word class has not been explored cross-linguistically; however, a significant literature within the language sciences has investigated related concepts.

While theoretical linguistics has focused on a distinction between content and function words, psycholinguistics has focused on semantic dimensions like imageability, concreteness, and strength of perceptual experience. Measures of these dimensions have relied on subjective, decontextualized human judgements, but nevertheless predict processing differences between word classes, such as asymmetries in the processing of nouns and verbs in certain aphasias (Bird et al., 2003; Dubé et al., 2014; Lin et al., 2022). Because we operationalize meaning as images, notions such as imageability seem especially related to our groundedness measure. However, as discussed in Section 5.4, these concepts differ from our measure in that informativity is not a major factor in their definition. For example, while both "zebra" and "woman" are highly concrete nouns, the former has higher groundedness on average, because although both are often strongly associated with an image, "zebra" is more informative/surprising, especially if the image is unavailable—thus, the image adds more information in that case.

As shown by the prior example, our measure is also closely related to another concept widely studied in computational psycholinguistics: *surprisal*. Like our groundedness measure, surprisal has an intuitive link to contentfulness from an information theoretic perspective, and has been extensively studied in relation to processing difficulty (Hale, 2001; Levy, 2008; Smith and Levy, 2013; Wilcox et al., 2023; Staub, Forthcoming). However, surprisal entangles formal and functional information in language. As such, cross-linguistic comparisons based on surprisal are challenging, since form is language specific (Park et al., 2021). We aim to focus on information due to language *function*, separated from form. Surprisal must also encode grammatical uncertainty (alternative ways of expressing the same meaning like "knight" and "cavalier"), as opposed to surprisal due only to what meanings are being expressed. Our image captioning model quantifies how many bits of information remain after the meaning is known. Our measure then quantifies how much of the LM surprisal is explained by the meaning (image).

3 Method

In this section, we define a token's *groundedness*, and show how we can use this to estimate the mutual information between parts of speech and representations of meaning. Let the set of word types in a language be $\mathcal{W}$. We assume a model of the data generation process where given a meaning m, a sentence is constructed by iteratively sampling a word $w_t \in \mathcal{W}$ conditioned on m and previous words $\mathbf{w}_{<t}$. As mentioned previously, the groundedness of a token is given by its pointwise mutual information (PMI) with the meaning.

$$\text{PMI}(w_t; m \mid \mathbf{w}_{<t}) = \log \frac{p(w_t \mid m, \mathbf{w}_{<t})}{p(w_t \mid \mathbf{w}_{<t})} \quad (1)$$

As we cannot access the true meaning m, we must approximate it with a proxy. A good proxy for m should be language-neutral, and will make estimating the probabilities in Equation 1 straightforward across languages. In this work, we focus on *images* as a language-neutral representation of meaning. Images capture rich, language-independent information about the world state described by an image, and have proved useful as a method for aligning meanings across languages (Rajendran et al., 2016; Gella et al., 2017; Mohammadshahi et al., 2019; Wu et al., 2022). Further, a major strength of images as a meaning representation is that estimating both quantities in Equation 1 becomes straightforward with neural models: $p_{\boldsymbol{\phi}}(w_t \mid m, \mathbf{w}_{<t})$ corresponds to the probability of the token under an image captioning model, while $p_{\boldsymbol{\theta}}(w_t \mid \mathbf{w}_{<t})$ corresponds to its probability under a language model.

Using images as a representation of meaning does have some implications for our approach. For instance, verbs, which usually denote events and are more temporally unstable (Givon, 1984) than other parts of speech, may be less grounded than with a different meaning representation, such as